Complete monotonicity of the image-charge interaction in a grounded slab: a heat-kernel proof and a Lean 4 formalisation

Xavier Callens
SocrateAI Research
https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids

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Abstract

Two pairs of opposite charges (two "excitons") interact, inside a slab bounded by two grounded perfect conductors, through the Dirichlet Green function of the Laplacian. We show that the resulting pair interaction $U(r)$, with r the lateral distance, is *completely monotone in* r^2: $U(r) = \int_0^\infty e^{-r^2\sigma}\mu(\sigma)\,d\sigma$ with a bounded non-negative density μ. The proof is a one-line consequence of the product structure of the Dirichlet heat kernel of the slab and of the positive definiteness of its transverse factor; the same argument covers any laterally homogeneous layered dielectric. A Lean 4/Mathlib formalisation (52 theorems, no `sorry`, standard axioms) proves the grounded-plane case by a different route, via Poisson summation. The statement is elementary and may well be known; we ask for that to be checked. We say precisely how it connects to universal-optimality theorems, and that the planar statement we would need there is not available in a refereed form.

1 Setting and statement

Let $h_1, h_2 > 0$, $\ell = h_1 + h_2$, $P = 2\ell$, and let $\Omega = \mathbb{R}^2 \times (-h_2, h_1)$ be the slab between grounded perfect conductors at $z = h_1$ and $z = -h_2$ (vacuum inside; layered media in §3). Write $G(x, z; x', z')$ for the Dirichlet Green function of $-\Delta$ on Ω, and $K = 4\pi G$, so that $K = 1/R$ in free space. By the method of images, with $R_\pm$ the distances to the charge and image charges,

$$K\big((x,z),(x',z')\big) = \sum_{n\in\mathbb{Z}} \Big[\frac{1}{\sqrt{r^2+(z-z'-Pn)^2}} - \frac{1}{\sqrt{r^2+(z+z'-2h_1-Pn)^2}}\Big], \qquad r = |x-x'|, \tag{1}$$

where the series converges absolutely, the two terms of each pair cancelling to $O(n^{-2})$.

An exciton consists of an electron (charge -1) in the plane $z = z_e$ and a hole ($+1$) in the plane $z = z_h$ at the same lateral position; the interaction energy of two excitons at lateral distance r, in units of e^2/ε and without r-independent self-image terms, is

$$U(r) = K_{ee}(r) + K_{hh}(r) - 2K_{eh}(r), \qquad K_{ab}(r) = K\big((x,z_a),(x',z_b)\big), \tag{2}$$

which has $U(r) \sim 2/r$ at 0 when $z_e \neq z_h$ lie in the open slab (and $U \equiv 0$ for $z_e = z_h$). The formula (1) makes sense for all real z, z' as the odd periodic extension; the Lean file proves the statement in that generality.

Theorem 1. *For all $z_e, z_h \in [-h_2, h_1]$ and all $h_1, h_2 > 0$ there is a measurable function $\mu : (0,\infty) \to [0,\infty)$, continuous and bounded, with*

$$U(r) = \int_0^\infty e^{-r^2\sigma}\,\mu(\sigma)\,d\sigma \qquad (r > 0).$$

In particular $t \mapsto U(\sqrt{t})$ is completely monotone on $(0,\infty)$: it is C^∞ and $(-1)^k \frac{d^k}{dt^k} U(\sqrt{t}) \geq 0$ for all k. Explicitly, with $\lambda_m = (m\pi/\ell)^2$ and $\varphi_m(z) = \sin\big(m\pi(z+h_2)/\ell\big)$,

$$\mu(\sigma) = \frac{4}{P\sigma} \sum_{m\geq 1} e^{-\lambda_m/(4\sigma)} \big(\varphi_m(z_e) - \varphi_m(z_h)\big)^2 \ \geq 0. \tag{3}$$

2 Proof via the Dirichlet heat kernel

Step 1 (product structure). The Dirichlet Laplacian on Ω is $-\Delta_x \otimes 1 + 1 \otimes A$ with $A = -\partial_z^2$ on $(-h_2, h_1)$ with Dirichlet conditions; the two summands act on different variables and commute, so the heat kernel of Ω is $p_s(x,z;x',z') = (4\pi s)^{-1} e^{-|x-x'|^2/4s}\, q_s(z,z')$, where q_s is the kernel of e^{-sA}. By the method of images

$$q_s(z,z') = \sum_{n\in\mathbb{Z}} \big[g_s(z-z'-Pn) - g_s(z+z'-2h_1-Pn)\big], \qquad g_s(u) = (4\pi s)^{-1/2} e^{-u^2/4s},$$

and also $q_s(z,z') = \frac{2}{\ell}\sum_{m\geq 1} e^{-s\lambda_m}\varphi_m(z)\varphi_m(z')$.

Step 2 (Laplace representation). $(-\Delta)^{-1} = \int_0^\infty e^{s\Delta}\, ds$ on L^2 with Dirichlet conditions, so $G = \int_0^\infty p_s\, ds$ and

$$K(r;z,z') = 4\pi G = \int_0^\infty \frac{1}{s}\, e^{-r^2/4s}\, q_s(z,z')\, ds = \int_0^\infty e^{-r^2\sigma}\, \frac{q_{1/4\sigma}(z,z')}{\sigma}\, d\sigma \qquad (\sigma = 1/4s).$$

(The integral converges for $r > 0$ because q_s is bounded by $g_s(0)$ and decays like $e^{-s\lambda_1}$ as $s \to \infty$.) Hence $U(r) = \int_0^\infty e^{-r^2\sigma}\mu(\sigma)\, d\sigma$ with $\mu(\sigma) = \sigma^{-1} S_{1/4\sigma}$ and

$$S_s := q_s(z_e,z_e) + q_s(z_h,z_h) - 2\, q_s(z_e,z_h). \tag{4}$$

Step 3 (positivity, the only point). By the semigroup property and symmetry of q, $q_s(z,z') = \int q_{s/2}(z,w)\, q_{s/2}(w,z')\, dw$, hence

$$S_s = \int_{-h_2}^{h_1} \big(q_{s/2}(z_e,w) - q_{s/2}(z_h,w)\big)^2\, dw \ \geq 0.$$

In the eigenfunction expansion this is $S_s = \frac{2}{\ell}\sum_m e^{-s\lambda_m}(\varphi_m(z_e) - \varphi_m(z_h))^2$, which with $s = 1/4\sigma$ gives (3). (Numerically, (1)–(3) agree to 10^{-12} at $r = 0.3, 1, 2.5$ (symmetric truncation $|n| \leq 2\times 10^5$) for $h_1 = 1, h_2 = 1.5, z_e = 0.3, z_h = -0.4$: `python/gated_note_check.py`.) (Neutrality of the exciton is what makes (4) a square: the non-neutral combination $K_{ee} - 2K_{eh}$ can be negative mode by mode; the Lean file contains this as a negative control.)

Step 4 (regularity, boundedness). Take $z_e \neq z_h$. As $\sigma \to 0$ every term of (3) carries $e^{-\lambda_m/4\sigma}$, so $\mu(\sigma) \to 0$. As $\sigma \to \infty$ ($s \to 0$), $S_s = 2g_s(0) + O(e^{-c/s})$ because the cross terms $q_s(z_e,z_h)$ and the image terms are exponentially small in $1/s$ for points at positive distance from each other and from the planes (for points on a plane the image terms vanish with the Dirichlet condition), hence $\mu(\sigma) \sim 2(\pi\sigma)^{-1/2}$: this is the $2/r$ singularity of U at 0, and it is integrable against $e^{-r^2\sigma}$ for every $r > 0$. μ is continuous and bounded on $(0,\infty)$ (the Lean file proves a bound by the Poisson form for $\sigma \geq 1$ and by the first term of the series for $\sigma \leq 1$). Differentiation under the integral, justified by $\sigma^k e^{-t\sigma} \leq C_{k,t_0} e^{-t_0\sigma/2}$ for $t \geq t_0$, gives $(-1)^k \frac{d^k}{dt^k}\int e^{-t\sigma}\mu = \int \sigma^k e^{-t\sigma}\mu \geq 0$ (the easy direction of Bernstein's theorem). □

Remark 1. Since $e^{-r^2\sigma}$ is a Gaussian in $x \in \mathbb{R}^d$ for every d, the representation exhibits U as a non-negative mixture of Gaussians. It is not a positive-definite *function* ($U(0) = \infty$), but it is a positive-definite distribution on $\mathbb{R}^d$ for $d \geq 2$ (its Fourier transform is positive and finite for $k \neq 0$; for $d = 1$, $2/|x|$ is not locally integrable). This is Bernstein's easy direction combined with Gaussians, not Schoenberg's theorem.

3 Layered dielectrics (not formalised)

Proposition 1. *Let $\varepsilon : (-h_2, h_1) \to [\varepsilon_{\min}, \varepsilon_{\max}] \subset (0,\infty)$ be measurable (a stack of homogeneous layers, or a uniaxial pair $\varepsilon_\parallel(z), \varepsilon_\perp(z)$, for which the lateral Laplacian carries $\varepsilon_\parallel/\varepsilon_\perp$ and the argument below is unchanged up to a rescaling of r), with grounded conductors at the two ends. Then (2) (with $K = 4\pi G$, G the Green function of $-\nabla\cdot\varepsilon\nabla$) satisfies $U(r) = \sum_{m\ge1} c_m\,(\psi_m(z_e) - \psi_m(z_h))^2 K_0(\sqrt{\lambda_m}\,r)$ with $c_m \ge 0$ (in units of e^2), where (λ_m, ψ_m), ψ_m normalised in $L^2(\varepsilon\,dz)$, are the Dirichlet eigenpairs of $-\varepsilon^{-1}\partial_z\varepsilon\partial_z$ in $L^2(\varepsilon\,dz)$. Since $K_0(\sqrt{\lambda t}) = \frac12\int_0^\infty e^{-ts}e^{-\lambda/4s}\,ds/s$, U is completely monotone in r^2.*

Proof. (i) The form $a(u,v) = \int \varepsilon u'v'\,dz$ on $H^1_0(-h_2,h_1)$ is closed, non-negative and has compact resolvent on $L^2(\varepsilon\,dz)$ (bounded ε, bounded below by $\varepsilon_{\min}$); its operator is $A = -\varepsilon^{-1}\partial_z\varepsilon\partial_z$, with discrete spectrum $0 < \lambda_1 \le \lambda_2 \le \dots$ and eigenfunctions ψ_m orthonormal in $L^2(\varepsilon\,dz)$. (ii) The potential of a unit charge at $(0,z')$ solves $-\nabla\cdot\varepsilon\nabla G = \delta$, i.e. $(-\Delta_x + A)G = \delta_x \otimes \delta_{z'}/\varepsilon(z')$; since e^{-sA} has a kernel $q_s(z,w)$ with respect to $\varepsilon\,dw$, and $\int q_s(z,w)\,\delta_{z'}(w)\varepsilon(w)^{-1}\varepsilon(w)\,dw = q_s(z,z')$, we get $G = \int_0^\infty p_s^{\mathbb{R}^2}(x)\,q_s(z,z')\,ds$ exactly as before (the heat semigroup is a tensor product because the two operators act on different variables). (iii) $q_s(z,z') = \sum_m e^{-s\lambda_m}\psi_m(z)\psi_m(z')$ is symmetric and satisfies $q_s = \int q_{s/2}\,q_{s/2}\,\varepsilon\,dw$, so $S_s = \int (q_{s/2}(z_e,w) - q_{s/2}(z_h,w))^2\varepsilon\,dw = \sum_m e^{-s\lambda_m}(\psi_m(z_e) - \psi_m(z_h))^2 \ge 0$. (iv) Integrability of $s^{-1}e^{-r^2/4s}S_s$ at $s\to0$ follows from $q_s(z,z) \le Cs^{-1/2}$ (Nash/ultracontractivity in one dimension), and at $s\to\infty$ from $q_s \le Ce^{-\lambda_1 s}$. All terms are non-negative, so by Tonelli the sum and the s-integral commute, and $\int_0^\infty s^{-1}e^{-r^2/4s}e^{-s\lambda}\,ds = 2K_0(\sqrt{\lambda}\,r)$ gives the expansion with $c_m = 2$ (the factor 4π of $K = 4\pi G$ cancels the $1/4\pi s$ of the lateral heat kernel); for the grounded planes with constant ε, $\psi_m = \sqrt{2/\ell}\,\varphi_m$ and this is $U = \frac{4}{\ell}\sum_m(\varphi_m(z_e) - \varphi_m(z_h))^2 K_0(\sqrt{\lambda_m}\,r)$, which agrees with the image series numerically. □

What this does not cover. Finite screening by the gates (non-ideal metals), lateral inhomogeneity, and quantum or polarisation effects are outside the statement; for them the product structure fails.

4 The Lean 4 formalisation

`exploration/exciton/lean/GatedKernel.lean` (1083 lines, 52 theorems; Lean 4.34.1, Mathlib v4.34.1; `import Mathlib` only). It proves Theorem 1 for the grounded planes by the *Fourier* route, not by the heat-kernel argument above: the image sums are Gaussian lattice sums, Poisson summation turns them into $\sum_m e^{-\pi^2m^2/(P^2s)}\cos(\cdot)$, and the combination (4) becomes a sum of $S_m = 2(1-\cos(b-a))(1+\cos(a+b)) = 2(\sin a - \sin b)^2$, which is the same quantity as $(\varphi_m(z_e) - \varphi_m(z_h))^2$ in (3).

declaration	content
`trig_core_sq`	$2 - \cos 2a - \cos 2b - 2\cos(b-a) + 2\cos(a+b) = 2(\sin a - \sin b)^2$
`gGreen_top`, `gGreen_bottom`	the Gaussian lattice Green function vanishes on both planes (bijective re-indexing of image offsets)
`gsum_poisson`	(sum over $m \in \mathbb{Z}$) $\sum_n e^{-s(c-Pn)^2} = \frac{\sqrt{\pi/s}}{P}\sum_m e^{-\pi^2m^2/(P^2s)}\cos(2\pi mc/P)$, *derived* from Mathlib's Gaussian/Jacobi-theta results
`Sexc_nonneg`	$S(s) \ge 0$ for all $s > 0$, $h_1 + h_2 > 0$, all real z_e, z_h
`muExc_bounded`, `muExc_continuousOn`	μ is bounded and continuous on $(0,\infty)$
`lap_admissible`	Laplace transform of a non-negative density with finite $\int e^{-st}\mu$ is smooth with $(-1)^r g^{(r)} \ge 0$ (easy Bernstein)
`Uimg_eq_lap`, `imageSeries_admissible`	the image series equals the Laplace transform, hence is `AdmissiblePotential`
`unneutral_not_nonneg`	negative control: dropping V_{hh} breaks positivity

All 17 `#print axioms` outputs are `[propext, Classical.choice, Quot.sound]`; there is no `sorry`, `axiom`, or `native_decide`. The only hypotheses are $0 < s, t$ and $0 < h_1 + h_2$. An independent verification (separate instance) compiled the file and recomputed the kernel by a direct signed image sum with $N = 4000$ (agreement 10^{-9} at three radii, and with the Laplace integral to 10^{-12}). Definition used: `AdmissiblePotential g` := g is C^∞ on $(0, \infty)$, $g \geq 0$, and $(-1)^r\, g^{(r)} \geq 0$ for all r (a verbatim copy of the definition in the upstream repository [1]). It is satisfied by $g = 0$, so non-triviality is a separate statement ($U \geq 0$ follows from $\mu \geq 0$; strict positivity for $z_e \neq z_h$ is not stated in the file). The Mathlib version pinned here has `AbsolutelyMonotoneOn` but no completely monotone class and no Laplace transform of measures, so the file carries its own.

5 Relation to universal optimality

Cohn and Kumar defined universally optimal configurations on spheres as those minimising the energy for *every* completely monotone function of squared distance [2], with the Euclidean version as their conjecture; E_8 and the Leech lattice were proved universally optimal in dimensions 8 and 24 by Cohn, Kumar, Miller, Radchenko and Viazovska [3]. The triangular lattice in dimension 2 is conjectured to be universally optimal in the same sense. A claimed proof of that planar statement is publicly available [1], but it is AI-generated and **has not been refereed to our knowledge**; we do not use it as a proved fact. What Theorem 1 supplies is exactly the hypothesis of that statement:

Proposition 2 (unconditional, lattices)**.** *Among Bravais lattices $\Lambda \subset \mathbb{R}^2$ of fixed density, the triangular lattice minimises $\sum_{0 \neq v \in \Lambda} U(|v|)$ for every z_e, z_h, h_1, h_2.*

Proof. By Theorem 1 and Tonelli, $\sum_v U(|v|) = \int_0^\infty \mu(\sigma)\, \Theta_\Lambda(\sigma)\, d\sigma$ with $\Theta_\Lambda(\sigma) = \sum_{v \neq 0} e^{-\sigma |v|^2}$, and $\mu \geq 0$. For every $\sigma > 0$ and every fixed density the triangular lattice minimises $\Theta_\Lambda(\sigma)$ among lattices (Rankin, Cassels, Ennola, Diananda; see Montgomery, Glasgow Math. J. **30** (1988) 75–85; references from memory). Integrate. □

This uses no AI-generated input. What remains conditional is the extension from lattices to *all* periodic configurations.

Proposition 3 (conditional)**.** If *the triangular lattice minimises, among periodic configurations of given density in $\mathbb{R}^2$, the energy per particle for every $f(r^2)$ with f completely monotone (and a convergence condition at 0 and ∞),* then *it does so for the gated kernel U of Theorem 1 for every z_e, z_h, h_1, h_2, since $U(\sqrt{t})$ is completely monotone and decays exponentially.*

This is a statement about periodic, bulk, classical configurations of the ideal-metal model. It says nothing about open boundaries, quantum states, or driven-dissipative dynamics, and a hexagonal pattern in an experiment does not single out complete monotonicity (the triangular lattice is the ground state of the Coulomb, Yukawa, dipolar and Gaussian-mixture potentials of this type, but not of all monotone repulsions, e.g. soft-core ones).

6 What we ask the reader

1. Is Theorem 1 known? It looks like the statement that a Dirichlet Green function in a product domain is a non-negative mixture of 2D heat kernels, applied to a neutral charge combination; a pointer to the literature would settle priority. (The modal expansion of Proposition 1 in K_0 modes is the classical parallel-plate expansion; what we believe is new is only the neutral-square positivity argument, the lattice proposition and the formalisation.)
2. Is Proposition 1 correct as stated (measurability/regularity of ε, endpoints, the weight $\varepsilon\, dz$)?

3. Is the status of the planar universal-optimality statement known to the reader, and is the conditional form above the right way to state the consequence?

4. Is the Lean statement a faithful encoding of (1)–(2)? (The kernel is the combination of six lattice sums `Uimg`; the verifier checked it numerically, not by a second formal proof.)

References

[1] OpenAI, *Universal optimality of the triangular lattice*, https://github.com/openai/math, family 090 (2026); AI-generated, unrefereed.

[2] H. Cohn and A. Kumar, *Universally optimal distribution of points on spheres*, J. Amer. Math. Soc. **20** (2007), 99–148.

[3] H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, Ann. of Math. **196** (2022), 983–1082.